

**Electrical and Computer Engineering Department
University of Puerto Rico at Mayagüez**



**COMMUNICATION THEORY I – MCA05
WHITE PAPER ON DIGITAL COMMUNICATIONS**

By

<SXXXGPYY>

For

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MCA05 – Due: Fri., Dec. 04, before 11:50 PM (MIDNITE)

WHITE PAPER ON DIGITAL COMMUNICATIONS

The objective of this **MCA05** is for each student group to write a [white paper](#), with and appendix and references, with a minimum of five pages (not including the reference page and the appendix) describing in a concise manner the essential and fundamental features of the selected topic. Each group must select a topic from the five topics provided in the list below. The document must be written in Microsoft Word, with the use of the equation editor. The document must be presented in .docx and .pdf formats, using the name "**INEL4301_SYYY_GPZZ_MCA05**" as its name. The appendix must include a paragraph of each member in each group describing in concrete details his/her contribution to the paper. The reference page must follow the [IEEE Reference Style](#).

01.- 5G Network Overview

Reference Paper: [Overview of 5G](#)
Reference Book: [5G Networks](#)
Reference Video: [Part 1](#), [Part 2](#), & [Part 3](#).

02.- Sensor Networks

Reference Paper: [Performance Analysis](#)
Reference Book: [Sensor Networks](#)
Reference Video: [WSN](#), [Libelium](#), [802.15.4 STANDARD](#).

03.- Underwater Communication

Reference Paper: [Platform for Underwater Comm.](#)
Reference Book: [Underwater Acoustic Communication](#)
Reference Video: [HOW?](#), [UAC](#), [High-Speed UW Comm.](#)

04.- Smart Antenna for Wireless Communication

Reference Paper: [Micro-Strip Patch Antenna Array](#)
Reference Book: [Smart Antenna Systems](#)
Reference Video: [Smart Antennas](#), [HOW?](#), [Beamforming](#).

05.- LEO Satellite Communications

Reference Paper: [LEO Coverage Analysis](#)
Reference Book: [LEO Satellite Design](#)
Reference Video: [HOW?](#), [Starlink](#), [Competitors](#), [LEO Intr.](#)